Ultimate Basic



Current version: **1.6.5**

A modern BASIC-like language that compiles directly to 6502 machine code for the **Commodore 64** and **Commodore 64 Ultimate**. It produces `.prg` files that run in VICE or on real hardware, and can also build `.d64` disk images.

Ultimate Basic looks like classic BASIC but compiles ahead of time — no interpreter, no line numbers required. It adds typed variables (`int` / `word` / `float` / `string` /arrays), subroutines and functions, structured control flow, and direct, high-level access to the C64's hardware: bitmap and block graphics, sprites, SID sound and music, raster/CIA/NMI interrupts, REU transfers, disk I/O, CRT cartridge export, and inline 6502 assembly.

Hires bitmap drawing (`plot` , `line` , `rect` , `circle` , `paint`) picks its color from `color pen c` — a persistent foreground color stamped into each touched cell. Multicolor bitmap mode has its own shape commands — `mplot` , `mline` , `mrect` , and `mcircle` — where the trailing `color` argument selects the 2-bit color source (0–3).

Sprite support includes hardware collision registers, side-effect-free software AABB tests with `box_hit()` , and frame selection from consecutive sprite animation data. Use `sprite_frame id, base, frame` inside an animation loop to select successive images stored in 64-byte slots; it changes the sprite image only and does not animate or move the sprite automatically.
Character tile maps can be embedded from self-describing `.ubmap` files, rendered as 40×25 viewports, queried and modified at runtime, in normal or multicolor text mode.
Koala Painter images can also be validated, embedded, shown in multicolor bitmap mode, and hidden again with `koala load` , `koala show` , and `koala hide` .

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Build

```
cargo build --release      # binary: target/release/ub
cargo test                 # unit + integration tests
cargo install cargo-deb cargo-generate-rpm
```

```
cargo deb                      # Debian/Ubuntu package: target/debian/*.deb
cargo generate-rpm             # Fedora/RHEL package: target/generate-rpm/*.rpm
```

The Debian and RPM commands package the release binary as `/usr/bin/ub` and include the README and manual under `/usr/share/doc/ultimate-basic`. The same packages are also built automatically by the `linux-packages` GitHub Actions job on pushes to `main` and can be downloaded from its workflow artifacts.

Usage

```
ub build demo.ub -o demo.prg          # compile to a .prg (prints a memory map)
ub build demo.ub -v                   # also print ZP layout + hex dump
ub build demo.ub --debug              # also write .sym, .dbg and .vs symbols
ub build demo.ub --asm                # also write a readable 6502 codegen listing
ub build demo.ub --d64 disk.d64       # also build a .d64 disk image
ub build demo.ub --d64 disk.d64 --add music.prg   # embed extra files in the .d64
```

Option	Description
`-o, --output <file>`	Output `.prg` (default: `<input>.prg`)
`-v, --verbose`	Print full zero-page layout and a hex dump
`--no-stub`	Skip the BASIC `SYS` stub (code loads at `$0801`)
`--debug`	Also produce KickAssembler `.sym`, C64Debugger `.dbg`, and VICE `.vs` files
`--asm`	Also produce a readable 6502 codegen listing as `<output>.asm`
`--d64 [file]`	Also produce a `.d64` (default: `<output>.d64`)
`--add <file>`	Add an extra file to the `.d64` (repeatable)
`--explicit`	Require a `:type` annotation on every `var`, `sub`-param, and `fn`-param

With `--debug`, the compiler writes three files beside the program: an importable KickAssembler `.sym`, a C64Debugger/RetroDebugger `.dbg`, and a VICE monitor `.vs`. They contain the program boundaries, variables, arrays, subroutines, and BASIC labels. The `.dbg` export currently provides address symbols but not source-line stepping data.

With `--asm`, the compiler writes a readable assembly listing beside the program. Unlike a plain disassembly of the finished PRG, it is built from code-generation metadata and retains UB statement comments, zero-page variable names, array/subroutine/BASIC-label symbols, generated branch/JMP/JSR labels, instruction addresses, and emitted machine-code bytes. Compiler-generated helper routines are included in the same listing.
The output uses KickAssembler syntax and can be assembled again. Known data regions—such as `data`, maps, sprite/character definitions, lookup tables, SID/Koala payloads, and `incbin` content—are emitted as `.byte` blocks instead of being mistaken for instructions.

What's new in 1.6.5

Bug fix release: **16-bit address expressions.**

- `peek($07F8 + k)`, `poke $07F8 + k, v`, `peek16` / `poke16` and the address and length operands of `fill`, `memcopy`, `drawmem`, `save` and `reu` are computed in 16 bits. Before, a constant plus a byte variable was added in 8 bits and the constant's high byte was dropped (`$07F8 + k` became `$00F8 + k`). `constant + byte` now compiles to a single `LDA $07F8,Y` / `STA $07F8,Y`.
- `poke ptr + i, v` with a `word` pointer is a full 16-bit add as well.
- `peek(k)` with a byte variable reads memory at `$00kk` (it returned `k` itself).
- `poke16 addr, base + k` writes the full 16-bit value.

What's new in 1.6.4

The three big items of `ROADMAP.md`:

- **Real floating point:** `DIM x AS SINGLE` / `AS DOUBLE` are 5-byte floating-point numbers as in C64 BASIC (~9 digits, ±1.7·10³⁸): `+ - * /`, `^`, `SQR`, `SIN`, `COS`, `TAN`, `ATN`, `EXP`, `LOG`, `INT`, `FIX`, `CINT`, `ABS`, `SGN`, literals like `1E6` / `.5`, comparisons, `FOR … STEP 0.1`, arrays, `AS DOUBLE` parameters and functions, `PRINT` / `STR$` / `VAL` / `INPUT` in C64 BASIC format. The runtime is generated by the compiler (no BASIC ROM calls) and added only when used. `float` stays the fast Q8.8 type. See `examples/float_demo.ub`.
- **Variables in RAM:** a program whose variables do not fit into zero page no longer fails with `out of zero page` — the least-used variables move to RAM automatically.
- **12 KB of arrays:** when arrays need more than 4 KB, they use `$A000-$CFFF` (RAM under the BASIC ROM, banked out while the program runs).
- Also: `%0101` binary literals; a leading minus binds to its operand (`-5 - 3` = -8, it was -2); `print 2.5` prints `2.50` (it printed the raw Q8.8 value 640); arrays declared inside subs and blocks get their own memory (they all overlapped at `$C000`).

What's new in 1.6.3

QBasic-style syntax — programs written the way BASIC books show them now compile. The existing `var` / `sub` / `fn` / `end` syntax is unchanged, and the two styles can be mixed.

- **`DIM name AS type [= value]`** — `BYTE` (8-bit), `INTEGER` (16-bit), `SINGLE` / `DOUBLE` (Q8.8 fixed point in 1.6.3 — real floating point since 1.6.4), `STRING`; several names (`DIM a, b AS INTEGER`) and mixed lists (`DIM n AS INTEGER = 1000, s AS STRING = "C64"`). Numbers start at 0, strings at `""`.
- **`DIM name(u1, u2, …) AS type`** — arrays of every type with any number of dimensions; each bound is the highest index (`DIM tabla(2, 3) AS INTEGER` is 3 × 4). Index with `tabla(r, c)` or `tabla[r, c]`. String arrays start with `""` elements.
- **`FUNCTION Negyzet(szam AS INTEGER) AS INTEGER … END FUNCTION`** — return a value by assigning to the function name; `EXIT FUNCTION` returns early.
- **`SUB Koszont(nev AS STRING) … END SUB`**, `EXIT SUB`; call it as `Koszont "Zsolt"`, `Koszont("Zsolt")` or `CALL Koszont("Zsolt")`.

- **`END IF`**, **`END SELECT`** accepted; **`PRINT a; b; c`** joins items (trailing `;` = no newline); `:` separates statements as before.
- Accented letters in strings print as their base letter (`Üdvözöllek` → `Udvozollek`) — the C64 charset has no accents.
- **More QBasic:** `IF x = 3` / `<>`, single-line `IF … THEN … ELSE …`, `ELSEIF`, `DO … LOOP` (WHILE / UNTIL at either end), `WHILE … WEND`, `SELECT CASE` with lists, `TO` ranges, `IS > n` and `CASE ELSE`, `EXIT DO / FOR`, `END`, and `'` comments.
- **Real runtime strings:** `s = "Pont: " + n + "!"`, `t = s` copies, `LEFT$`, `RIGHT$`, `MID$`, `LEN` of any string expression, string `=` / `<>`, 16-bit `STR$` / `VAL`.
- **Signed `INTEGER`** (−32768 … 32767): printing, comparisons, division, `ABS`, loops through zero, negative literals anywhere (`a < -10`). `WORD` stays unsigned.
- **`INPUT`** into INTEGER / WORD (16-bit), DOUBLE (`12.5`), array elements, and `INPUT "prompt"; x`.
- **Exact `DOUBLE ÷ DOUBLE`**; recursive subs/functions are now a compile error instead of silently wrong.
- New examples: `examples/dim_demo.ub`, `examples/qbasic_demo.ub`. Deferred larger work (ROM floating point, more variables, more array space) is described in `ROADMAP.md`.

Fixes: `s = "TEXT"` on a string variable, a string literal passed to a `string` parameter, both bytes of `word` / `float` parameters, and printing / 16-bit arithmetic with `word`-array elements (`print t[i]` printed 0, `w = t[i] + 5` used only the low byte). Also:

- a `for` loop with a 16-bit counter wrapped at 256 (`for x = 0 to 319` never got past 255, e.g. in `wide_x_demo.ub`);
- arrays larger than 256 bytes indexed by a variable wrote into the wrong element;
- `print f(30)` of a 16-bit function printed only the low byte;
- `a * b` of two 16-bit values used only the low byte of `b` (300 * 300 gave 13200);
- the `map_view` unit test overflowed in debug builds.

What's new in 1.6.2

Richer `tune` instruments — the same features as the Visual Assembler SID editor:

- **`imod id, vibdelay, vibspeed, vibdepth, pwmspeed, pwmmin, pwmmax`** — instrument vibrato (a delayed triangle LFO scaled to the note's semitone) and pulse width modulation bouncing between two widths.
- **`ifilt id, cutend, sweep, pingpong`** — filter cutoff sweep toward `cutend`, optionally back and forth: resonant filter sweeps. The voice that starts a filter-routed instrument owns the filter; routing is per voice now.
- **`igate id, gatetimer, hardrestart, firstwave`** — gate timer and hard restart before the next note, and the waveform of a note's first frame (e.g. `$09`).
- **`itab id, speed, loop, "41 +4", ".. +7", "81 =60"`** — wave / arpeggio tables for chords, drums and waveform sequences.
- Ring modulation and sync bits in `inst` ctrl, 3 OFF in `modevol` bit 7.

The player runs these every frame and is checked frame by frame against the SID editor's engine (`tests/tune_player.rs`).

What's new in 1.6.1

Smooth scrolling maps for games:

- **`map view px, py`** shows the loaded map at a pixel position: XSCROLL / YSCROLL for the fine part, two screen buffers (`$0400` / `$3C00`) swapped at char steps, the next char cell built ahead of time and color RAM copied in step with the raster beam — pixel-smooth and tear-free in every direction. Call it once per frame instead of `wait raster`; sprites keep working (their pointers are copied to both buffers).
- **`map load "level.bin"`** reads a VisualAssembler Level Editor project directly and composes its screens into one map (up to 6×10 screens, 255×255 cells).
- **`map load "file", addr`** places the map data at a fixed address instead of inline in the code, like `incbin "file", addr` — e.g. `$4000` , which keeps `$3C00-$3FFF` free for `map view` .
- New example: `examples/level_scroll_demo.ub` (joystick-driven level with a player sprite).

Fixes: a byte product or `shl` assigned to a `word` inside a sum now keeps 16 bits (`py = cy * 8 + fy` was truncated to 8 bits), and a constant multiplier above 255 (`cy * 300`) is no longer cut to 8 bits.

What's new in 1.6.0

Faster, smaller machine code — the code generator was reworked (`codegen/opt.rs`). Existing programs compile unchanged and simply run faster:

- **Conditions** (`if` / `while` / `until` / `select`) compile to a `CMP` followed directly by a branch instead of building a 0/1 value and testing it again (`if x == 10` shrank from 24 to 9 bytes). `x > n` becomes `x >= n+1` , `and` / `or` of comparisons short-circuit, `while` loops test at the bottom, and `select` compares the variable in place.
- **Direct operands:** constants and byte variables are used as `ADC #n` / `CMP zp` operands instead of being copied to scratch zero page; `poke` to a constant or `word` address stores directly; identity operations (`x + 0` , `x * 1` , …) vanish.
- **Multiply / divide / modulo:** `*` by a constant is shifts / shift-add (`x * 8` → 3× `ASL`), otherwise a shift-add loop of at most 8 rounds. `/` and `mod` by a power of two are `LSR` / `AND` , otherwise an 8-round shift-subtract loop — no longer proportional to the quotient (`255 / 1` used to take 255 rounds).
- **Increments:** `x = x + 1` / `x - 1` / `± 2` become `INC` / `DEC` (word variables: 16-bit `INC` / `DEC` for ±1).
- **Arrays** with a variable index use `LDA/STA $C000,Y` instead of a zero-page pointer.
- **`for` loops** are rotated (test at the bottom, no `JMP` per iteration); constant limits and steps are immediates and use no zero page.
- **A-register reuse:** a statement that starts by loading the variable the previous statement just stored skips the redundant `LDA` .

Behaviour fixes that came with it:

- **Division by zero no longer hangs:** `x / 0 = 255` , `x mod 0 = x` .
- **`for i = 0 to 255`** runs 256 times and stops (the 8-bit wrap-around is detected in both directions) instead of looping forever.

New test infrastructure: a full 6502 emulator (`tests/common/cpu6502.rs`), semantic tests that run compiled programs and compare results with Rust (`tests/codegen_semantics.rs`), code-shape / cycle tests (`tests/codegen_opt.rs`), and an opt-in differential fuzzer against a reference compiler (`tests/differential_fuzz.rs`).

What's new in 1.5.9

- **`chain "FILE" [, device]`** loads another program over the running one and RUNs it, as if `LOAD"FILE",8,1` + `RUN` had been typed — the missing piece for a boot / title program that shows a picture and then starts the game. A small position-independent loader is copied to the cassette buffer (`$033C`), the I/O is reset (IOINIT, RESTOR, VIC IRQs off), BASIC is re-initialised after the load (so Exomizer `sfx` crunched boot programs work too) and the program is started. If the file cannot be loaded, execution continues after `chain`.

```
koala load "title.koa"
koala show
k = getch()
chain "GAME"
print "LOAD ERROR"
```

- **Fix: `load` secondary address.** `load "FILE"` used secondary address 0 with X/Y = 0, so the file was loaded to `$0000`; `load "FILE", addr` ignored `addr`. Now `load "FILE"` loads to the file's own address and `load "FILE", addr` to `addr`.

- **Fix: `--d64` images with more than one file.** Directory entries were written 30 bytes apart instead of 32, so every file after the first had a broken directory entry (a `--add`ed file could not be loaded). The disk ID / DOS type were also in the wrong BAM bytes, data only used tracks 1–17 (a bigger disk crashed the build), and sectors were written without the 1541 interleave. Images are now laid out as the 1541 DOS writes them (tracks 17→1, then 19→35, interleave 10), and a full disk is a clean build error.

What's new in 1.5.8

- **Initialised arrays: `data <array>: v, …`** fill a declared `array` / `array_word` with constant values. Several lines continue where the previous one stopped, word arrays take 16-bit values (stored lo, hi), and the table is indexed like any array (`sinus[i]`). The values are stored after the code and copied into the array once at program start, right after the arrays are zeroed. These lines do not affect the `read` stream; more values than the array holds, or an undeclared name, is a compile-time error. This is what the Visual Assembler Curve Editor's "Export to UltimateBasic" generates.

```
var sinus = array(8)
data sinus: 128, 218, 255, 218
```

```
data sinus: 128, 37, 0, 37
sprite 0, x, sinus[i]
```

What's new in 1.5.7

New language elements:

- **`charset on`** / **`charset off`** switch the VIC-II to the RAM character set at `charset addr` (and back to the ROM set). Together with `chardef` this is a complete custom-font workflow. A bad base address, or program code that reaches into the set, is a compile-time error.
- **`org addr`** continues the subroutine code at a later address and leaves the gap free for data, so a big program can live around a charset, sprite data or a font.
- **`incbin "file", addr, skip [, len]`** skips a `.prg` load address (or any header) and can take just a slice of a file. `incbin ..., addr` may now target an `org` gap.
- **`sfx channel, freq, frames [, wave]`** is a non-blocking SID note: it returns at once and the envelope fades the note out, so it can be used inside a game loop (`sound` blocks).
- **`tune ... end`** describes a 3-voice SID tracker tune (instruments, order list and pattern rows such as `"C-4 01 V24"`) directly in the source. The compiler builds a self-contained player at compile time - it uses no zero page - and defines `sid_init` / `sid_play`, so `music play|stop|pause|resume` work as with `load sid`. Vibrato, slide, note cut and speed changes are supported. This is what the Visual Assembler SID editor's "Export to UltimateBasic" generates.

```
tune
  speed 6
  inst 0, $41, $09, $F0, $0800, $0064, $27, $1F
  order 0
  pat 0, 0, "C-4 00", "...", "E-4 00 V24", "...", "G-4 00", "... .. C00"
end
music play
```

```
charset $3800
incbin "font.bin", $3800          # data goes into the gap that `org` leaves
...
org $4000                         # (between two subs) code continues above the charset
charset on                        # the VIC-II now draws from $3800
sfx 1, $0500, 8, 128              # a short noise burst; the program keeps running
```

Fixes:

- **`music play`** stopped after the first tick: its CIA1 IRQ wrapper acknowledged the timer by writing `$01` to `$DC0D` (which clears the interrupt mask) instead of reading it.
- **Inline `asm { }`** dropped the addressing mode of label operands, so `LDA table,X`, `LDA table,Y` and `JMP (label)` were assembled as plain absolute instructions.
- **`sound`** lasted half as long as requested and crashed for durations other than 0 (a wrong branch patch). Now it waits on raster line 200 like `delay`.
- **`word` products** (`score += 300 * level`, `w = 100 * l + 900`) are now computed in 16 bits; they were truncated to 8 bits.
- **`print` of float expressions** (`print a / 10`) prints `N.DD` instead of the raw Q8.8 integer.
- **Arrays start zeroed** (RAM at power-up holds garbage).
- **Zero page overflow** is a compile-time error (`out of zero page`), and sequential `for` loops share their limit/step bytes instead of exhausting the permanent area.
- The manual no longer claims `%` binary literals (the lexer never supported them) and the `charset` example uses the right `$D018` value.

What's new in 1.5.6

- Added **`color pen` and multicolor shapes**. `color pen c` sets a persistent hires drawing color (0–15) that `plot`, `line`, `rect`, `circle`, and `paint` stamp into the foreground nibble of every cell they touch (the background nibble is preserved; default white). For multicolor bitmap mode, `mline`, `mrect`, and `mcircle` join `mplot`, each taking a trailing 2-bit `color` (0–3) source:

```
graphics on
gcls
color pen 2
circle 160, 100, 40      # red circle

graphics on multi
gcls
mcircle 80, 100, 40, 1   # multicolor circle
mline 0, 0, 159, 199, 3
```

  The multicolor shapes reuse the same Bresenham/midpoint routines as their hires counterparts, plotting each pixel through `mplot`.

What's new in 1.5.5

- Added **Magic Desk CRT export**. Building with an output filename ending in `.crt` now writes a Commodore 64 cartridge image instead of a raw PRG:

```
ub build demo.ub -o demo.crt
```

The cartridge uses the same banked Magic Desk layout as VisualAssembler's CRT export: an 8×8K type-19 image with a boot loader in bank 0 and the compiled PRG payload copied into RAM on startup. Existing `.prg` and `.d64` output paths are unchanged.

- Added integration-test coverage for the CRT packer so the cartridge header, bank layout, payload copy, and boot stub stay stable.
- Added **`type ... endtype` structs**. Define a fixed layout of `int` (1 byte) or `word` (2 bytes) fields, then allocate an array of instances with the same syntax as any other array:

```
type TEntity
  var x:  int
  var y:  int
  var hp: word
endtype

const N = 8
var enemies: TEntity = array(N)   # N × 4 = 32 bytes at $C000
enemies[0].x  = 5
enemies[i].hp = enemies[i].hp + 1
```

Constant indices fold to `LDA/STA absolute`; variable indices emit a shift-and-add `idx * elem_size` multiplier followed by `(ptr),Y` access. See `examples/type_demo.ub`. Struct-typed sub/fn parameters, default field values, and nested struct fields are not yet implemented.

- **Count-down `for` / `loop`** with negative constant `step` now works correctly. Previously the exit test was unsigned-only, so `for i = 20 to 0 step -2` either ran zero iterations or looped forever depending on whether it wrapped past 0. The compiler now picks the exit-branch encoding from the sign of the constant `step`, and emits an extra post-increment `BCS` check that catches the 8-bit underflow when counting down to (or through) 0.
- A `for` / `loop` with default `step +1` where `from > to` — e.g. `for i = 10 to 1` — used to skip the body silently. It is now a compile-time error directing you to `step -1`:

```
for-loop: from (10) > to (1) with default step +1 loops 0 times
```

```
    — use 'step -1' to count down
```

- Added the **`--explicit`** CLI flag. When passed to `ub build`, every `var`, `sub`-param, and `fn`-param declaration must carry a `:type` annotation; arrays and `const` are unaffected. The flag is a build-time switch, so nothing in the source changes to opt in or out. See `examples/explicit_demo.ub` and `examples/explicit_errors_demo.ub`.
- Added `examples/countdown_demo.ub` and `examples/countdown_errors_demo.ub`, plus new integration tests that both check the emitted branch encoding and run the loop on the test CPU to confirm it terminates.

What's new in 1.5.3

- Added **multi-dimensional arrays**. Declare an array with a comma-separated dimension list and index it the same way; storage is row-major and the total size is the product of the dimensions:

```
var grid = array(8, 8)      # 8×8 = 64 bytes
grid[r, c] = 42             # row-major: base + r*8 + c
var v = grid[r, c]
var wm = array_word(4, 4)   # word (16-bit) elements work too
```

  Any number of dimensions is supported, dimensions may be `const`s, all-constant subscripts fold to a direct address at compile time, and a single subscript into a multi-dimensional array is still allowed as flat/linear access.
- Fixed PETSCII string encoding of `[`, `]`, and `^`, which previously printed as `?`. They now emit their correct C64 codes (`$5B`, `$5D`, `$5E`) in both the uppercase and lowercase charset modes.
- Added `examples/array2d_demo.ub` and integration tests for multi-dimensional arrays and the bracket encoding.

What's new in 1.5.2

- Added `ub build <file.ub> --asm` and `<file>.asm` codegen listings.
- Listings retain UB statement boundaries and compiler symbols instead of producing only a bare post-build disassembly.
- Zero-page variables and arrays are emitted as named assembler constants.
- BASIC labels, subroutines, relative branches, and generated in-program JMP/JSR targets receive readable labels.

- Every instruction includes its C64 address and generated machine-code bytes, making the output useful for inspection, debugging, optimization, and comparison with the PRG.
- Compiler helpers use descriptive `ub_helper_*` labels, while known embedded data is emitted with named `.byte` regions in reassemblable KickAssembler syntax.
- Added listing tests and documentation in the README, manual, CLI help, release notes, and compiler developer guide.

```
## Documentation

The complete language and CLI reference lives in **[MANUAL.md](MANUAL.md)** — variables and
types, operators, control flow, subroutines/functions, graphics (bitmap, double-buffered,
block), sprites, sound and SID music, interrupts, REU, disk I/O, string/math functions, and
inline assembly.

Release history is in [whatnews.txt](whatnews.txt).

## Examples

Ready-to-build demos are in [`examples/`](examples/) — bitmap and block graphics, sprite
multiplexing, plasma and orbit effects, a flicker-free double-buffered 3D cube
(`cube_demo.ub`), REU stash/fetch, SID music playback, scrollers, QBasic-style
`DIM` / `FUNCTION` / `SUB` (`dim_demo.ub`, `qbasic_demo.ub`), floating point
(`float_demo.ub`), and more.
```